Homework Assignment 1

Math 292, Spring 2026

Rutgers University

Problem 1:

(a) Find the general solution of

$$tx'(t) = x(t) + 3t^2.$$

Solution:

1. First, divide through by t and rearrange

$$x'(t) - \tfrac{1}{t}x(t) = 3t$$

2. Solve for the integrating factor

$$\begin{aligned} e^{-\int \frac{1}{t}dt} &= e^{-\ln|t|} \\ &= \frac{1}{t} \end{aligned}$$

3. Multiply both sides by integrating factor

$$\begin{aligned} \frac{x'(t)}{t} - \frac{1}{t^2} &= 3 \\ \frac{d}{dt}(\frac{1}{t}x(t)) &= 3 \end{aligned}$$

4. Integrate both sides w.r.t. t and simplify for general solution

$$\begin{aligned} \int \frac{d}{dt}(\frac{1}{t}x(t)) &= \int 3dt \\ \frac{x(t)}{t} &= 3t + C \mid C \in \mathbb{R} \\ x(t) &= 3t^2 + Ct \end{aligned}$$

(b) Find the flow transformation $\Phi_{t,t_0}(x)$ specified by this equation.

Solution:

1. Plug in t_0 into the general solution and solve for C

$$\begin{aligned} x(t_0) &= 3t_0^2 + Ct_0 \\ x(t_0) - 3t_0^2 &= Ct_0 \\ \frac{x(t_0)}{t_0} - 3t_0 &= C \end{aligned}$$

2. Now using this C, you can find the flow transformation

$$\begin{aligned} x(t) &= 3t^2 + t(\frac{x(t_0)}{t_0} - 3t_0) \\ \Phi_{t_0,t}(x(t)) &= 3t^2 + t(\frac{x(t_0)}{t_0} - 3t_0) \end{aligned}$$

(c) Find the particular solution $x(t)$ that satisfies $x(1) = 2$.

Solution:

1. Solve for the C constant using the general solution

$$\begin{aligned} x(1) &= 3(1^2) + C(1) \\ 2 &= 3 + C \\ C &= -1 \end{aligned}$$

2. Plug C into general solution to find the particular solution for this initial condition

$$x(t) = 3t^2 - t$$

Problem 2:

(a) Find the general solution of

$$x'(t) + \frac{1}{3}x(t) = e^t x^4(t).$$

Solution:

1. Notice that this solution is in the of Bernoulli equation, thus we must use substitution to reduce it to this to a first order linear differential equation

$$\begin{aligned} \text{Let } u &= x(t)^{-3} \\ u' &= -3x'(t)x(t)^{-4} \\ -\frac{1}{3}u' &= x'(t)x(t)^4 \end{aligned}$$

2. Divide through equation by $x(t)^4$

$$x'(t)x^{-4} + \frac{1}{3}x^{-3} = e^t$$

3. Now, rewrite the equation in terms of u and t only, then simplify

$$\begin{aligned} -\frac{u'}{3} + \frac{1}{3}u &= e^t \\ u' - u &= -3e^t \end{aligned}$$

4. We now have a first order linear ode, so we solve for the integrating factor

$$\begin{aligned} \text{Let } \mu &= e^{-\int 1dt} \\ &= e^{-t} \end{aligned}$$

5. Multiply both sides of the equation by this integrating factor

$$\begin{aligned} e^{-t}u' - e^{-t}u &= -3e^{-t}e^t \\ \frac{d}{dt}(e^{-t}u) &= -3 \end{aligned}$$

6. Now integrate both sides w.r.t. to t

$$\begin{aligned} \int \frac{d}{dt}(e^{-t}u) &= \int -3dt \\ e^{-t}u &= -3t + C \end{aligned}$$

7. Isolate u

$$u = -3e^t t + e^t C$$

8. Substitute x back in for u

$$\begin{aligned} x(t)^{-3} &= e^t(-3t + C) \\ x(t) &= e^{-\frac{t}{3}}(C - 3t)^{-\frac{1}{3}} \end{aligned}$$

(b) Find the particular solution $x(t)$ that satisfies $x(1) = 2$. Over what time interval $t \in (a, b)$ is the solution a continuously differentiable function?

Solution:

1. Solve for C using this initial condition and the general solution we attained above

$$x(1) = e^{-\frac{1}{3}}(C - 3(1))^{-\frac{1}{3}}$$
$$\frac{1}{8} = e(C - 3)$$
$$C = \frac{1}{8e} + 3$$

2. Plug this C back into general solution and simplify

$$x(t) = e^{-\frac{t}{3}}(\frac{1}{8e} + 3 - 3t)^{-\frac{1}{3}}$$
$$x(t) = \frac{1}{e^{\frac{t}{3}}(\frac{1}{8}e^{-1} + 3 - 3t)^{\frac{1}{3}}}$$

3. As long as the denominator is nonzero, the solution will be a continuously differentiable function. Thus, we set the denominator to zero and solve for t. Notice that e cannot be zero since its strictly positive and thus we only care about what is in the parenthesis

$$0 = e^{\frac{t}{3}}(\frac{1}{8}e^{-1} + 3 - 3t)^{\frac{1}{3}}$$
$$0 = (\frac{1}{8}e^{-1} + 3 - 3t)$$
$$t = \frac{1}{24}e^{-1} + 1$$

4. From this, we can conclude that the solution "blows up" after $t = \frac{1}{24}e^{-1} + 1$ and since this is greater than our initial condition of t=1. Our valid time interval must be

$$t \in (-\infty, \frac{1}{24}e^{-1} + 1)$$

Problem 3:

(a) Find the general solution of

$$tx'(t) = tx^2(t) - x(t) - \frac{1}{t}$$

Solution:

1. First, isolate $x'(t)$ by dividing through by t

$$x'(t) = x^2(t) - \frac{x(t)}{t} - \frac{1}{t^2}$$

2. Notice that the equation is in the form of a Ricatti equation, so we plug in the $x_1(t) = ct^\alpha \mid c, \alpha \in \mathbb{R}$ to try to guess the particular solution.

$$-ct^{\alpha-1} = c^2t^{2\alpha} - ct^{\alpha-1} - t^{-2}$$

3. Set all the exponents equal to solve for α

$$\alpha - 1 = 2\alpha = -2\alpha = -1$$

4. Balancing the exponents, we see that α has to be -1, now we solve for c

$$\begin{aligned} -ct^{-2} &= c^2t^{-2} - ct^{-2} - t^{-2} \\ c &= -c^2 + c + 1 \\ c &= \pm 1 \end{aligned}$$

5. Choosing the positive c, we get the particular solution:

$$\begin{aligned} x_p(t) &= t^{-1} \\ x_p'(t) &= -t^{-2} \end{aligned}$$

6. Now consider the following Bernoulli equation

$$u' - (q + 2rx_p)u - ru^2 = 0$$

and let

$$\begin{aligned} x_p(t) &= t^{-1} \\ q &= -t^{-1} \\ r &= 1 \end{aligned}$$

7. Solve for $\mu = u^{1-n}$

$$\begin{aligned} \mu &= u^{-1} \\ \mu' &= -u'u^{-2} \end{aligned}$$

8. Divide through by x^2

$$u'u^{-2} - t^{-1}u^{-1} - 1 = 0$$

9. Substitute u with μ and rearrange

$$\begin{aligned} -\mu' - t^{-1}\mu - 1 &= 0 \\ \mu' + t^{-1}\mu &= -1 \end{aligned}$$

10. Now we have a first order linear ODE, next we solve for integrating factor

$$\begin{aligned} e^{\int t^{-1}dt} &= e^{\ln|t|} \\ &= t \end{aligned}$$

11. Multiply both sides by the integrating factor and integrate

$$\begin{aligned} \mu' t + \mu &= -t \\ \int \frac{d}{dt}(\mu t) &= -\int t dt \\ \mu t &= -\frac{t^2}{2} + C \\ \mu &= -\frac{t}{2} + \frac{C}{t} \end{aligned}$$

12. Substitute back into $x(t) = x_p(t) + \mu$, we have the final general solution

$$\begin{aligned} x(t) &= t^{-1} - \frac{t}{2} + \frac{C}{t} \\ &= t^{-1} + \frac{2t}{2C - t^-} \end{aligned}$$

(b) For any (x_0, t_0) with $t_0 > 0$, find the solution $x(t)$ of this equation that satisfies $x(t_0) = x_0$.

Solution:

1. We substitute $t = t_0$ in the general solution and solve for 2C

$$\begin{aligned} x_0 &= t_0^{-1} + \frac{2t_0}{2C - t_0^{-2}} \\ x_0 - t_0^{-1} &= \frac{2t_0}{2C - t_0^{-2}} \\ 2C - t_0^{-2} &= \frac{2t_0}{x_0 - t_0^{-1}} \\ 2C - t_0^{-2} &= \frac{2t_0^2}{x_0 t_0 - 1} \\ 2C &= \frac{2t_0^2}{x_0 t_0 - 1} + t_0^2 \\ 2C &= \frac{x_0 t_0^3 + t_0^2}{x_0 t_0 - 1} \end{aligned}$$

2. Plug back into the general solution and simplify for the final solution

$$\begin{aligned}\Phi_{t,t_0}(x) &= t^{-1} + \frac{2t}{\frac{x_0t_0^3+t_0^2}{x_0t_0-1} - t^2} \\ &= t^{-1} + \frac{2t(x_0t_0-1)}{t_0^2(x_0t_0+1) - t^2(x_0t_0-1)}\end{aligned}$$

(c) Write down a formula for the flow transformation Φ_{t,t_0} generated by this equation. Verify explicitly that $\Phi_{3,2}(\Phi_{2,1}(x)) = \Phi_{3,1}(x)$ for all x.

Solution:

1. Now, we explicitly check $\Phi_{3,1}(x)$ using the solution from part b

$$\begin{aligned}\Phi_{3,1}(x) &= 3^{-1} + \frac{2(3)(x(1)-1)}{(1)^2(x(1)+1) - (3)^2(x(1)-1)} \\ &= 3^{-1} + \frac{6x-6}{10-8x} \\ &= \frac{(10-8x)+3(6x-6)}{3(10-8x)} \\ &= \frac{5x-4}{15-12x}\end{aligned}$$

2. Then, we need to solve for $\Phi_{2,1}(x)$

$$\begin{aligned}\Phi_{2,1}(x) &= 2^{-1} + \frac{2*2(x-1)}{(x+1) - 2^2(x-1)} \\ &= 2^{-1} + \frac{4x-4}{5-3x} \\ &= \frac{5-3x+2(4x-4)}{2(5x-3)} \\ &= \frac{5x-3}{10-6x}\end{aligned}$$

3. We use the above solution and plug it in as the input for x_0 in $\Phi_{3,2}$

$$\begin{aligned}
\Phi_{3,2}(\Phi_{2,1}(x)) &= \frac{1}{3} + \frac{6(\frac{(5x-3)}{10-6x} - 1)}{4(2(\frac{(5x-3)}{10-6x}) + 1) - 3^2(2(\frac{(5x-3)}{10-6x}) - 1)} \\
&= \frac{1}{3} + \frac{6\frac{5x-3}{5-3x} - 6}{4(\frac{5x-3}{5-3x}) + 4 - 9(\frac{5x-3}{5-3x}) + 9} \\
&= \frac{1}{3} + \frac{6\frac{5x-3}{5-3x} - 6}{-5(\frac{5x-3}{5-3x}) + 13} \\
&= \frac{1}{3} + \frac{48x - 48}{80 - 64x} \\
&= \frac{1}{3} + \frac{3x - 3}{5 - 4x} \\
&= \frac{5 - 4x + 3(3x - 3)}{3(5x - 4)} \\
&= \frac{5x - 4}{15 - 12x}
\end{aligned}$$

4. This is indeed the same as the answer we got in step 1

Problem 4: Consider the equation

$$x'(t) = 2t\frac{t^2 + x(t)}{t^2 - x(t)} \tag{1}$$

This is not first order linear, Bernoulli, or Riccati, and it is not separable as it stands. But introducing a new variable, we can get a more amenable equation. Introduce $y(t)$ through $y(t) = \frac{x(t)}{t^2}$ for $t > 0$.

(a) Show that $x(t)$ solves (1) for $t > 0$ if and only if $y(t)$ solves a separable equation for $t > 0$.

Solution:

1. First, we let

$$y(t) = \frac{x(t)}{t^2} \mid t > 0$$

2. Rearrange and differentiate both sides w.r.t t

$$x(t) = y(t)t^2$$
$$x'(t) = y'(t)t^2 + 2ty(t)$$

3. Substitute what we got above back into the original ODE and solve for $y'(t)$

$$y'(t)t^2 + 2ty(t) = 2t\frac{t^2 + t^2y(t)}{t^2 - t^2y(t)}$$
$$y'(t)t^2 + 2ty(t) = 2t\frac{1 + y(t)}{1 - y(t)}$$
$$y'(t)t^2 = 2t(\frac{1 + y(t)}{1 - y(t)} - y(t))$$
$$y'(t)t^2 = 2t(\frac{1 + y^2(t)}{1 - y(t)})$$
$$y'(t) = \frac{2}{t}(\frac{1 + y^2(t)}{1 - y(t)})$$

4. This is now a separable differentiable equation, thus x(t) solves (1) iff y(t) solves the separable equation below for $t > 0$

$$dy(\frac{1 - y(t)}{1 + y^2(t)}) = \frac{2}{t}dt$$

(b) Find the solution of the separable equation with $y(1) = y_0$, $y_0 \neq -1$.

Solution:

1. Integrating both sides of the separable differentiable equation we got above

$$\int y'(t)(\frac{1-y(t)}{1+y^2(t)}) = \int \frac{1}{t}dt$$
$$\int \frac{1}{1+y^2}dy - \int \frac{y}{1+y^2}dy = 2\ln|t| + C$$
$$\arctan(y) - \frac{1}{2}\ln|1+y^2| = 2\ln|t| + C$$
$$\arctan(y) = \ln(t^2\sqrt{1+y^2}) + C$$

(c) Find the solution of (1) with $x(1) = x_0$, $x_0 \neq -1$.

Solution:

1. We just have to substitute x back in for y and simplify to get the general solution of (1)

$$\arctan(\frac{x}{t^2}) = \ln(t^2\sqrt{1+(\frac{x}{t^2})^2}) + C$$
$$\arctan(\frac{x}{t^2}) = \ln(t^2\sqrt{\frac{t^4+x^2}{t^4}}) + C$$
$$\arctan(\frac{x}{t^2}) = \ln(\sqrt{t^4+x^2}) + C$$

Problem 5: Find the general solution of the equation

$$tx''(t) = 1 + (x'(t))^2.$$

Solution:

1. We can use reduction of order and substitute $y = x'(t)$ so that the equation becomes

$$ty' = 1 + y^2$$

2. This can easily be rearranged into a separable differentiable equation and solved

$$\begin{aligned} ty' &= 1 + y^2 \\ \frac{1}{1+y^2}dy &= \frac{1}{t}dt \\ \arctan(y) &= \ln|t| + Cy \qquad\qquad = \tan(\ln|t| + C) \end{aligned}$$

3. Now we bring back the x' and integrate again to get x

$$\begin{aligned} \int x'dt &= \int \tan(\ln|t| + C)dt \\ x &= \int \tan(\ln|t| + C_1)dt + C_2 \mid \{C_1, C_2\} \in \mathbb{R} \end{aligned}$$